(xv) When 'n' equall successive discounts are given on M.R.P then

$$S.P = MRP \times (x)^n$$

[∴ x = after giving each equal successive discount resultant value in decimal]

Questions

Ex:-1:- [B.Q:-1]:- A milkman buys 2 goats for 30000. He sells 1st @ profit of 22% & 2nd @ loss of 8%. What is SP of 2nd if in the whole transaction no profit no loss?

Ans:- $CP_1 + CP_2 = 30000 \xrightarrow{①} 1.22CP_1 + 0.92CP_2$ [NoP/NoL]

$1.22CP_1 - CP_1 = CP_2 - 0.92CP_2 \Rightarrow 0.8CP_2 = 0.22CP_1$

$\Rightarrow CP_1 = \frac{8}{22}CP_2 = \frac{4}{11}CP_2 \quad -②$

② in ① $\Rightarrow \frac{4}{11}CP_2 + CP_2 = 30000 \Rightarrow 15CP_2 = 30000 \times 11.$

$\Rightarrow CP_2 = 22000$

But we want $SP_2 = 0.92CP_2 = 0.92 \times 22000 = \boxed{20240}$

Ex2:- [B.Q:2]:- Sum of CP's of 2 camels is 39000/-. Both the camels are sold @ profit of 20% & 40% respectively with their SP's being same. What is Diff b/w C.P's?

Ans:- $CP_1 + CP_2 = 39000 \quad -①$, $SP_1 = 1.2CP_1$, $SP_2 = 1.4CP_2$

$SP_1 = SP_2 \Rightarrow 1.2CP_1 = 1.4CP_2 \Rightarrow CP_1 = \frac{1.4}{1.2}CP_2 = \frac{7}{6}CP_2 \quad -②$

Sub ② in ①

$\Rightarrow \frac{7}{6}CP_2 + CP_2 = 39000 \Rightarrow \frac{13}{6}CP_2 = 39000$

$\Rightarrow CP_2 = 18000 \quad -③$

③ in ② $\Rightarrow \frac{7}{6}CP_2 = CP_1 = \frac{7}{6} \times 18000 = 21000$

Diff = 21000 − 18000 = $\boxed{3000}$